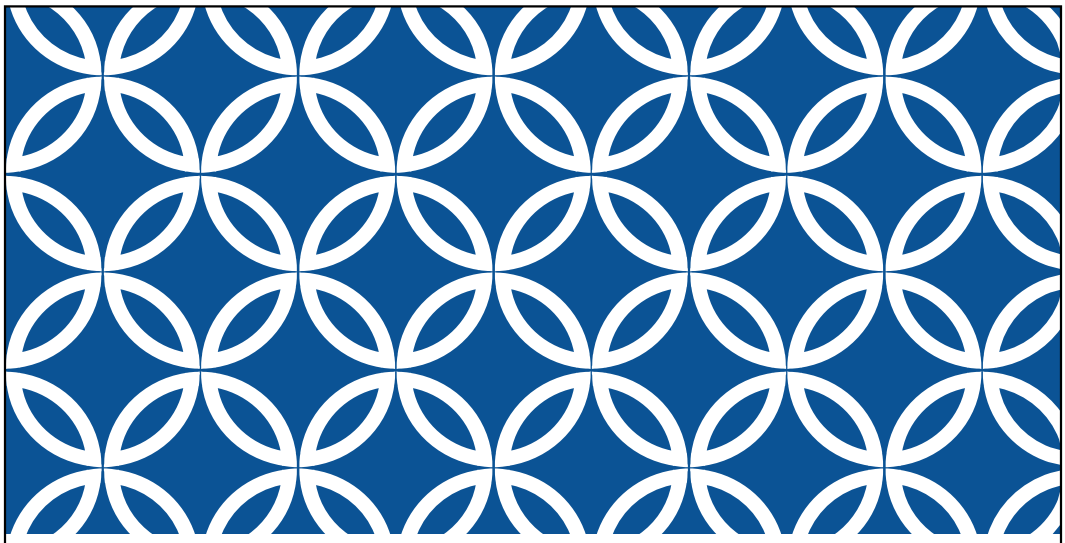


DOSAGE CALCULATION #2 | NUR3260

DIMENSIONAL ANALYSIS



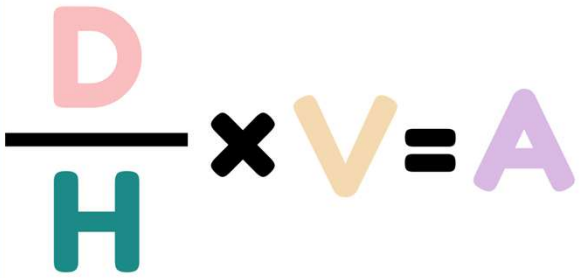
- What are you solving for? What units should the answer be in?
- Always start with the order or what you know.
- Find the conversion factor (what you have on hand)
- Make sure units match. If they don't, we need to convert!
- Line matching units up diagonally so they cross out
- Double check the answer to see if it is in the units that were asked for

FORMULA METHOD

Desired dose (ordered)

Have (available dose)

Volume (quantity)



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DIMENSIONAL ANALYSIS VS FORMULA

Ordered: 160mg Acetaminophen

Available: 80mg/ml Acetaminophen

How many milliliters to admin?

DA VS FORMULA WITH CONVERSION

Ordered: 200mcg Cytotec

Available: 0.2mg tablets

How many tablets to admin?

ORAL TABLETS/PILLS/CAPSULES



Most medications nurses give are given orally

Rounded to the nearest half or quarter **tablet**

The HCP ordered 650 mg of acetaminophen every 8 hours prn. The nurse has 325 mg tablets on hand. How many tablets will the nurse administer per dose? Round to the nearest half tablet.

$$650 \text{ mg} \times \frac{1 \text{ tab}}{325 \text{ mg}} = 2 \text{ tablets}$$

- What are you solving for? What units should the answer be in?- **tablets**
- Always start with the order or what you know- **650 mg**
- Find the conversion factor (what you have on hand)- **none needed here**
- Make sure units match. If they don't, we need to convert!- **mg's on top… mg's on bottom!**
- Double check the answer to see if it is in the units that were asked for- **answer should be in tablets.**

LIQUID MEDICATIONS

Liquid medications come in a variety of concentrations

- Injectable - Rarely more than 3 mL (*SAFETY*)
- Oral

The order reads 8 mg of ondansetron to be given IVP





LIQUID ORAL MEDICATIONS: PRACTICE!

The order reads 0.25 grams of liquid Motrin po. The bottle reads 100 mg/5 ml. How many milliliters do you administer (round to nearest whole number)?

INJECTABLE: PRACTICE!

Demerol 40 mg is ordered IM. The vial is labeled 50 mg/mL. How many mL do you prepare (round to nearest tenth spot)?

MORE PRACTICE

Ordered 250mcg

Available: 2.5mg per 2mL

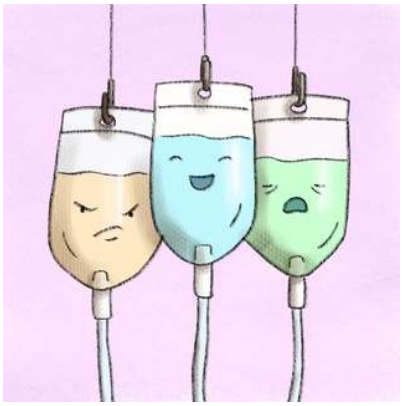
10/14/2026

Or convert 2.5 mg to mcg first then do D/HxV.

FLOW RATES

IV fluids are given two ways: by gravity or by a pump

Nurses use pumps more often because it is more accurate and safer



FLOW RATE BY GRAVITY



No pump is used! Nurse will count the drops in the drip chamber and adjust the flow rate with the roller clamp

Answer is in gtts/min = drops of fluid in the drip chamber per minute

Different tubing sets have different drop factors (gtts per mL)

Formula:

$$\frac{\text{Volume (ml)}}{\text{Time (mins)}} \times \text{Drop Factor (gtt/ml)}$$

Answer should be round to WHOLE number

Can't have a fraction of a drop

FLOW RATE BY PUMP

IV pumps are used frequently: The pump is in control of the flow rate

Nurse programs the pump

Answer is in ml/hr= milliliters administered to the patient in one hour

Formula:

$$\frac{\text{Volume (ml)}}{\text{Time (hrs)}}$$

Round where the problem tells you! Most pumps can have a tenth spot on IV rates



GRAVITY VS PUMP

Gravity

Key words to look for:

- Gravity
- Drops or gtts
- Drop factor or calibration factor

Formula:

$$\frac{\text{Volume (ml)}}{\text{Time (mins)}} \times \text{Drop Factor (gtt/ml)}$$

Rounding:

Whole number

Pump

Key words to look for:

- Pump
- ml/hr

Formula:

$$\frac{\text{Volume (ml)}}{\text{Time (hrs)}}$$

Rounding:

Whole number for maintenance fluids

Tenths place for medications

FLOW RATE: PRACTICE!

1000 ml of NS to run over 8 hours. The tubing has a drop factor of 10 gtts per ml. How many drops/minute should the nurse deliver?

Key Words: drop factor, gtts, drops

FLOW RATE: PRACTICE!

Infuse 1 liter (1000 mL) of NS over 8 hours. What is the flow rate in ml/hr?

FLOW RATE PRACTICE

Order: Infuse 2,500 mL of D5W over 24 hours (round to nearest whole number)

FLOW RATE: PRACTICE

Order: Infuse 500mg Ampicillin over 40 minutes.

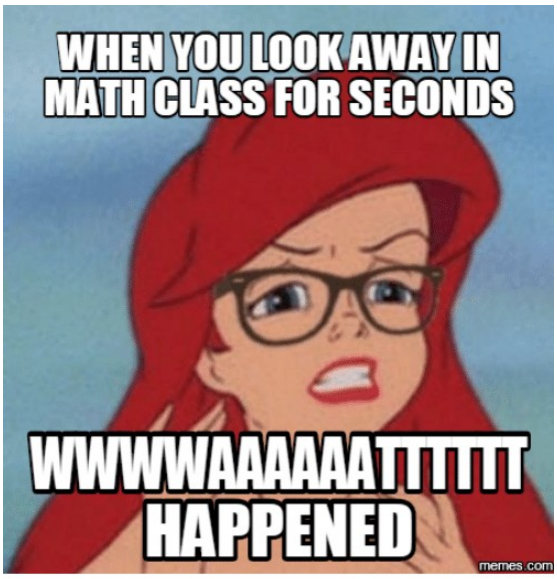
Available Ampicillin 300mg in 30mL

Grip factor 20 gtt/mL

RESOURCES

- https://nurseslabs.com/drug-dosage-calculations-practice-quiz/
  - Has a lot of practice problems and immediate rationales and demonstrations for how to complete
- https://www.dosagehelp.com
  - Can choose specific questions styles
- https://www.leveluprn.com/blogs/dosage-calculation
  - Videos on calculations

QUESTIONS?


WHEN YOU LOOK AWAY IN MATH CLASS FOR SECONDS
WWWWAAAAAATTTTTT HAPPENED
memes.com